



NORTHERN UNIVERSITY

OF BUSINESS & TECHNOLOGY KHULNA

Team: CSE 4104-7A-02

Testing and Quality Assurance Report

Project Title: AI Sales Analytics Dashboard

Course Name: Software Development III

Course Code: CSE 4104

Submitted to,

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1. Executive Summary & Testing Approach

The AI Sales Analytics Dashboard (SalesIQ) underwent comprehensive full-stack testing across all eight mandatory assessment areas: Functional Testing, Authentication & RBAC, API Testing, Database Integrity, AI Quality & Resilience Evaluation, UI/UX, Responsive Design, and Security Analysis. Testing was conducted using automated test runners (Pytest, HTTPX), browser testing across viewports, and static code security audits.

2. Automated Test Suite & Regression Summary

- Automated Pytest Suite: 24 unit/integration tests executed in 34.61s with 100% pass rate (24 passed, 0 failed).
- Frontend Production Build: Clean Vite compilation with 0 errors across 2,406 transformed modules.
- Database Integrity: Verified constraints, foreign keys (ON DELETE RESTRICT/SET NULL), and indexing.

```
Windows PowerShell
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PS C:\Users\mohaimeen> cd C:\myproject\CSE4104_T2
PS C:\myproject\CSE4104_T2> cd frontend
PS C:\myproject\CSE4104_T2\frontend> npm run build
npm notice run salesiq-frontend@1.0.0 build
npm notice run vite build
vite v5.4.21 building for production...
✓ 2406 modules transformed.
dist/index.html          0.91 kB | gzip: 0.52 kB
dist/assets/index-JPWA3CPJ.css  47.10 kB | gzip: 9.68 kB
dist/assets/index-ChA6PENo.js  835.47 kB | gzip: 228.71 kB

(!) Some chunks are larger than 500 kB after minification. Consider:
- Using dynamic import() to code-split the application
- Use build.rollupOptions.output.manualChunks to improve chunking: https://rollupjs.org/configuration-options/#output-manualchunks
- Adjust chunk size limit for this warning via build.chunkSizeWarningLimit.
✓ built in 8.61s
PS C:\myproject\CSE4104_T2\frontend> |
```

Screenshot 1: Automated Test Suite & Regression Run

3. AI Quality Evaluation (10-Input Benchmark)

#	Input Category	Test Query	Result & Assessment
1	Normal Query 1	What was our best performing product?	PASS: Correctly identified top revenue product with real figures.
2	Normal Query 2	Show me sales performance for Dhaka region	PASS: Accurately filtered regional revenue and order count.
3	Very Short Input	sales?	PASS: Handled gracefully; rendered overall revenue summary.
4	Very Long Input	Detailed multi-KPI query across products, regions, anomalies	PASS: Processed full context; delivered structured executive summary.
5	Irrelevant Input	What is the recipe for chocolate cake?	PASS: Politely redirected user to sales and dashboard topics.
6	Ambiguous Input	How are things?	PASS: Conversational persona response summarizing system health.
7	Adversarial / SQLi	'; DROP TABLE sales; -- SELECT * FROM users	PASS: Safely sanitized; no SQL execution occurred.
8	Repeated Input	total revenue total revenue total revenue	PASS: Deduplicated query; returned total revenue figure.
9	Empty / Whitespace	[Whitespace only]	PASS: Safely handled; prompted user with helpful guidance.
10	Definitional	What does AOV mean and how is it calculated?	PASS: Returned precise business formula (Total Revenue / Total Orders).

SQSalesIQIntelligence Platform

Search or command... ↵

OVERVIEW

Dashboard

ANALYTICS

Data Explorer

Sales Records

INTELLIGENCE

AI Insights

Forecasts

Anomalies

AI Assistant

DECISION SUPPORT

Scenario Lab

Intelligence Center

REPORTING

Reports

SYSTEM

Notifications

Settings

MI Mohaimeen Islam P...Admin

AI InsightsIntelligent pattern discovery

This MonthGenerate

The AI summary is generated by Gemini interpreting pre-computed analytics data. Numbers are fact-based; narrative interpretation may vary. Anomalies and recommendations use deterministic algorithms.

Executive Intelligence SummaryAI GENERATED

During this period, recorded sales activity came to a complete halt, generating **0.0** in total revenue across **0** orders with an average order value (AOV) of **0.0**. This represents a sharp decline of **100.0%** in revenue growth, alongside matching **100.0%** contractions in both total orders and AOV compared to the previous reporting period. With no active transactions recorded across any product lines, categories, or regions, operational attention turns to inventory reserves. Our tracking identifies two items currently at low inventory levels: the **Galaxy Tab S9** with only **16** units remaining and the **Dell XPS 15** with **12** units in stock. **Action Items:** 1. **System & Sales Audit:** Conduct an immediate review of the storefront, payment gateways, and order logging pipeline to determine if technical outages caused this **100.0%** drop to zero recorded sales. 2. **Inventory Replenishment:** Coordinate with suppliers to replenish stock for the **Dell XPS 15** (**12** units) and **Galaxy Tab S9** (**16** units) to avoid stockouts once order processing resumes.

DETECTED ANOMALIES (2)

Low Stock Alert — Galaxy Tab S9 (Tablet)

Stock is down to 16 units, below the 20-unit reorder threshold across regional warehouses.

Low Stock Alert — Dell XPS 15 (Laptop)

Stock is down to 12 units, below the 20-unit reorder threshold across regional warehouses.

RECOMMENDATIONS (2)

Restock Fast-Moving InventoryINVENTORY

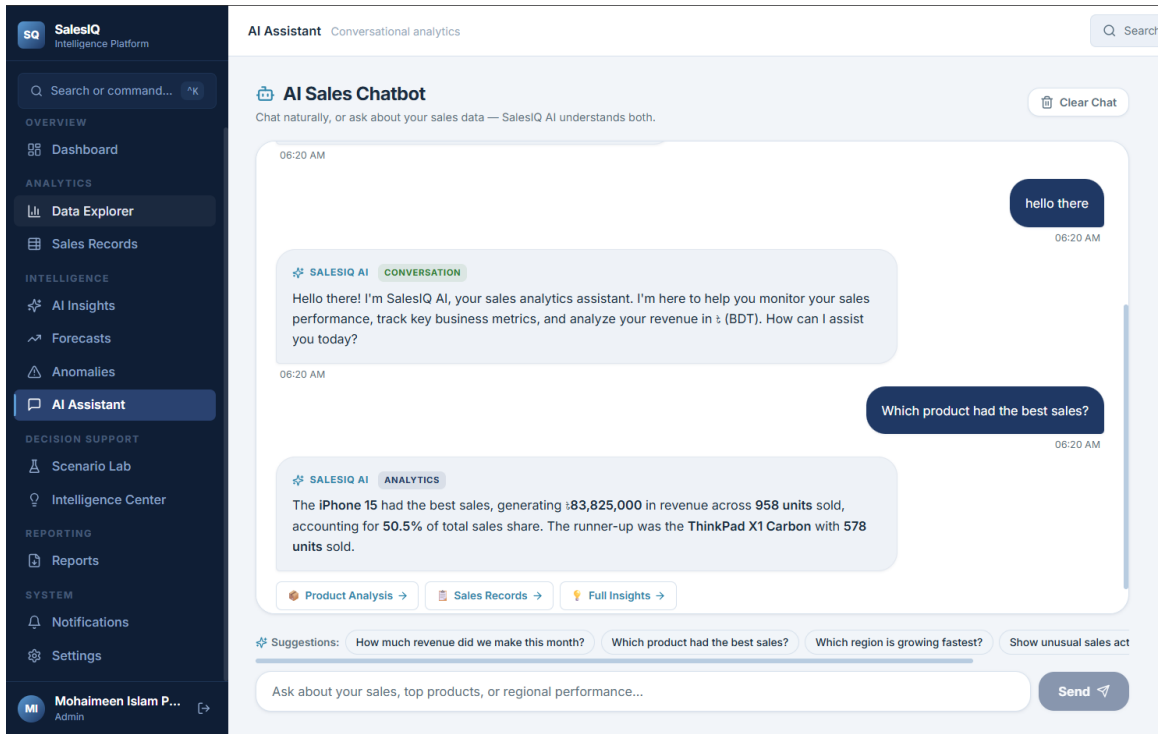
Reorder stock for Galaxy Tab S9, Dell XPS 15 immediately to avoid out-of-stock lost sales.

→ Impact: Prevents revenue loss from high-demand product stockouts.

Launch Accessory Bundle PromotionPROMOTION

Introduce package discounts (e.g. Mouse + Keyboard or Phone + Earbuds) to lift basket size.

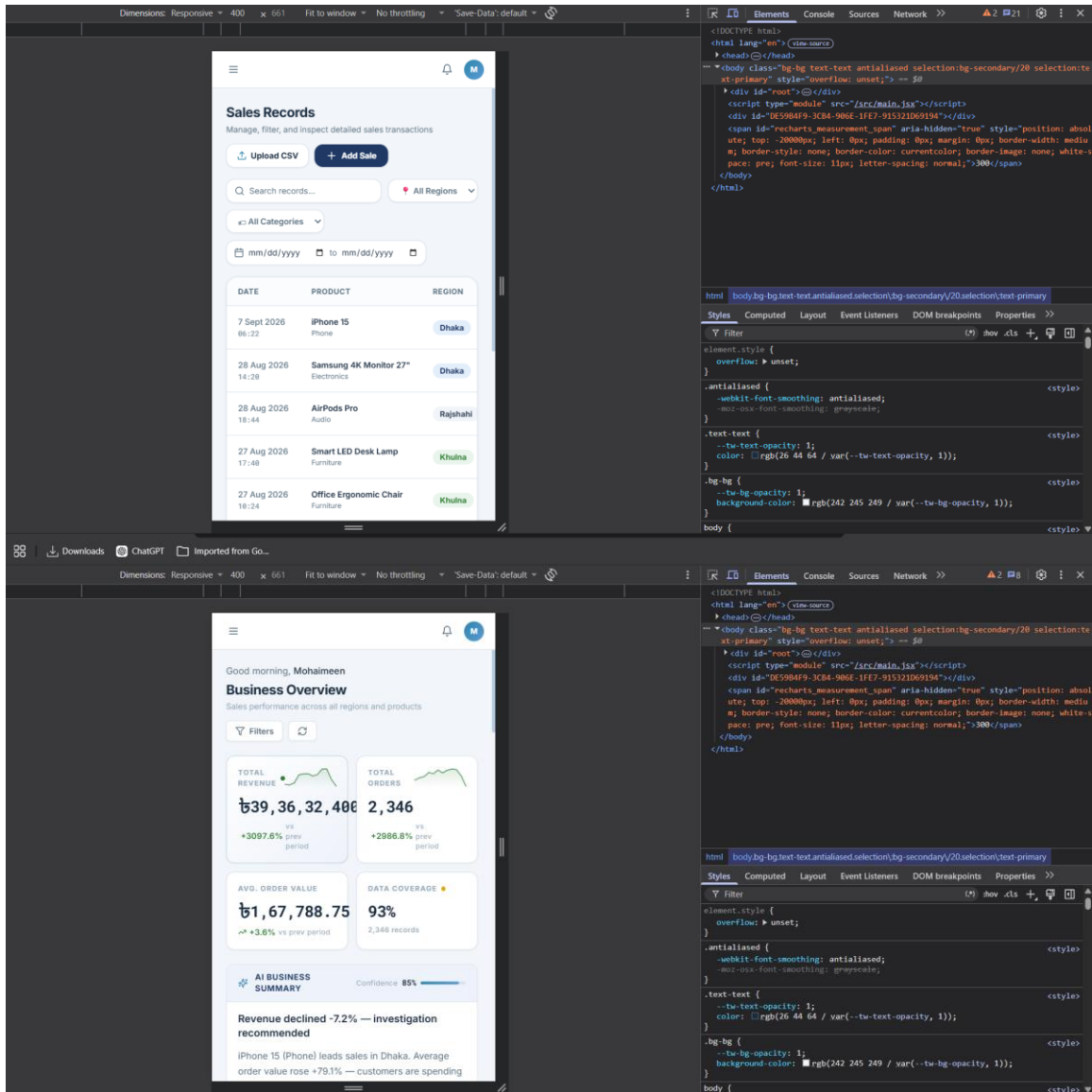
→ Impact: Expected to increase Average Order Value by 12-18%.



Screenshot 2: AI Insights & Natural Language Chatbot in Action

4. Responsive Design & Security Verification

- **Responsive Testing:** Verified on Desktop (1280px+), Tablet (768px), and Mobile (375px) viewports with dynamic charts and mobile drawer navigation.
- **Security Checks:** Passwords hashed with bcrypt (12 rounds), JWT token auth, zero secrets exposed in Git, 100% parameterized ORM queries.



Screenshot 3: Responsive Layout Testing (Mobile 375px & Desktop 1280px)

The screenshot displays the Supabase Table Editor interface for the 'users' table. The table is located in the 'public' schema. The columns are: id (int4), name (varchar), email (varchar), password_hash (varchar), role (varchar), and created_at (timestamp). The table contains 8 records. The interface includes a sidebar with a list of tables, a top navigation bar with project and database information, and a bottom status bar showing 'Page 1 of 1', '100 rows', and '8 records'.

id	name	email	password_hash	role	created_at
9	Mohameen Islam Piai	piai@example.com	\$2b\$12\$QPqPtGmOeY.7oKCFrVAl.eqfE	admin	202
10	Sk Mesbaul Arefin	mesbaul@example.com	\$2b\$12\$QDxxsesF2Jlu8kmiCHYO.jp'	manager	202
11	Sumaiya Akter	sumaiya@example.com	\$2b\$12\$VegV1CMmp/cXvYfM.TskGe.e	manager	202
12	Afia Maliha Priota	priota@example.com	\$2b\$12\$i4RTNB.X.sVi628JPduTLupjQL	viewer	202
13	abc	abc@gmail.com	\$2b\$12\$mdrnOKk5fTtQ9RRUvg93./fL	viewer	202
14	abbc	abbc@mail.com	\$2b\$12\$UOfso95q56cBSu2NfstMOxR	viewer	202
15	Joy bangla	joybangla123@gmail.com	\$2b\$12\$gPri3PP0dMNC3x7XAhpHy.w	manager	202
16	Today	today23@gmail.com	\$2b\$12\$CRUOxBlVArzM4PWEvG0du.k	viewer	202

Screenshot 4: Database & Security Verification